Tarun Trivedi

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Education

Carnegie Mellon University *Sept 2020 – Dec 2023*
BS in Applied Mathematics and Computer Science

- **Coursework:** Autonomous Agents, Machine Learning (Graduate), Deep Learning (Graduate), Natural Language Processing (Graduate), Computer Systems, Deep Reinforcement Learning (Graduate), Algorithms for Big Data, Computer Vision, Probability Theory (Honors), Linear Algebra (Honors), Game Theory, Statistical Inference
- **Extracurricular:** Dean's list, Carnegie Mellon student ambassador, All University Orchestra section leader, Jazz Band section leader, VP of CMU cycling team

École Polytechnique Fédérale de Lausanne *Dec 2022 – August 2023*
Visiting Student in Mathematics

- **Coursework:** Stochastic Processes, Martingales in Financial Mathematics (Graduate), Discrete Optimization (Graduate)
- **Research Assistant:** Lévy driven models for volatility pricing

Experience

ML Researcher *Chicago, IL*
Chicago Trading Company *Jan 2024 – Current*

- One of three researchers on a trading team that went from R&D to consistently profitable
- Building neural and recurrent networks to predict price movement in near dated index options
- Utilize ensemble methods of LLMs, neural networks, GBDTs, and feature learned linear models to predict price movements on short horizons
- Leverage internal cloud framework for running simulations and jobs to dump data for model fitting
- Write high-performance C++ strategy code for options taking and quoting, as well as for computing features for real-time trading with low latency
- Sklearn, lightgbm, pytorch, pandas, and numpy

Research Assistant *Pittsburgh, PA*
Carnegie Mellon University *Aug 2023 – Aug 2024*

- Research assistant at the RASL lab under the robotics institute at Carnegie Mellon
- Assisted Ph.D. student Roshni Kaushik in the following paper:
  Personalized Context-aware Multimodal Robot Feedback
- Implemented dynamic time warping algorithm to compare temporal angle time sequences of participants' completion of exercises
- Given the result of this algorithm, a robot was able to give feedback to participants on how to improve their exercise form

Teaching *Pittsburgh, PA*
Carnegie Mellon University *Jan 2022 – May 2022*

- Teaching assistant for Functional Programming, taught in the computer science department
- Led recitations, graded homeworks, and ran office hours for students
- Peer tutor for Linear Algebra, Differential Equations, and Intro to Artifical Intelligence

Projects

Side Projects

- March Madness Kaggle prediction competition (top 100)
- Training gpt-2 from scratch

- Playing around with MNIST dataset for mechanistic interpretability
- Stanford RNA 3D Folding Data Science Competition: Applying deep learning to an RNA sequence to predict its 3D structure
- Ribonanza RNA Folding Data Science Competition: Predicting reactivity of an RNA sequence to two chemical modifiers DMS and 2A3. Predicts reactivity down to every position in an RNA molecule.
- Single-cell multi-omics: Exploring using graph neural networks for cell fate prediction via performing clustering on sc RNA-seq and sc ATAC-seq data